EXPERIMENT – 02

AIM :- Convert the ER diagram into a relational schema. Write complete MySQL CREATE TABLE elements with primary keys, foreign keys, NOT NULL, ENSURE, DELETE, CASCADE and set null NULL constraints with sample data and demonstrate referential integrity

THEORY :- A relational schema converts this conceptual design into tables can be implemented in a relational database such as MySQL. Each entity generally becomes a table, attributes become columns and relationship are represented using primary and foreign keys. Constraints such as NOT NULL, UNIQUE and FOREIGN KEY maintain data accuracy and consistency.

INTEGRITY CONSTRAINTS :- Referential integrity ensures that a foreign key value must correspond to an existing primary key in the referenced table.

- ON DELETE CASCADE – deletes child records automatic
- ON DELETE SET NULL – Keeps the child record but set its foreign key to NULL.
- NOT NULL – prevents missing values.
- UNIQUE – prevents duplicate values.
- PRIMARY KEY – Uniquely identifies each record.

RELATIONAL SCHEMA :-

Customer (
- customer_id PK
- name
- email UNIQUE
- phone
- registered_ON
)

Address (
- address_id PK,
- customer_id FK,
- address_type,
- street,
- city,
- pincode,
- country
)

Seller (
- seller_id PK,
- seller_name,
- email UNIQUE
)

Category (
- category_id PK,
- category_name UNIQUE,
- description
)

Product (
- product_id PK,
- product_name,
- price,
- seller_id FK,
- category_id FK,
)

Orders (
- order_id PK,
- customer_id FK,
- order_date,
- status,
- total_amount
)

Order_Item (
- order_id PK/FK,
- product_id PK/FK,
- price,
- quantity
)

Payment (
- payment_id PK,
- order_id FK,
- transaction_id UNIQUE,
- payment_date,
- amount,
- payment_method,
- payment_status,
)

Delivery (
- delivery_id PK,
- order_id FK,
- address_id FK,
- delivery_date,
- delivery_status,
- tracking_no UNIQUE,
- delivery_partner
)

1 : N Relationship :- primary key of the '1' side is added as a foreign key to the N side.

M : N Relationship :- a new junction table is created containing the primary key of both participating entities.

1 : 1 Relationship :- A foreign key can be placed in preferably on the side with total participation.

SQL :-

```
CREATE DATABASE IF NOT EXISTS ecommerce_db;
USE ecommerce_db;
```

1) Customer

```
CREATE TABLE Customer (
        customer_id INT PRIMARY KEY,
        name VARCHAR (100) NOT NULL,
        email VARCHAR (150) NOT NULL, UNIQUE,
        phone VARCHAR (15) NOT NULL,
        registered_on DATE TIME NOT NULL
);
```

2) Address

```
CREATE TABLE Address (
        address_id INT PRIMARY KEY,
        customer_id INT NOT NULL,
        address_type VARCHAR (30) NOT NULL,
        Street VARCHAR (150) NOT NULL,
        city VARCHAR ( 50 ) NOT NULL,
        State VARCHAR ( 50 ) NOT NULL,
        pincode VARCHAR ( 10 ) NOT NULL,
        Country VARCHAR ( 50 ) NOT NULL, DEFAULT 'India',
```

```
    CONSTRAINT FK_address_customer
        FOREIGN KEY (customer_id)
        REFERENCES customer (customer_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);
```

3) SELLER

```
CREATE TABLE Seller (
    Seller_id INT PRIMARY KEY,
    Seller_name varchar (100) NOT NULL,
    email VARCHAR (150) NOT NULL UNIQUE
);
```

4) CATEGORY

```
CREATE TABLE category (
    category_id INT PRIMARY KEY
    category_name VARCHAR (100) NOT NULL UNIQUE,
    description VARCHAR (255)
);
```

5) PRODUCT

```
CREATE TABLE Product (
    product_id INT PRIMARY KEY,
    product_name VARCHAR (150) NOT NULL,
    price decimal (10, 2) NOT NULL,
    seller_id INT,
    category_id INT,
    CONSTRAINT chk_product_price,
        CHECK (price >= 0),
```

```
CONSTRAINT   FK_product_seller
        FORBIGN KBY (seller_id)
        RBFBRBNCBS Seller (seller_id)
        ON DBLBTB SBT NULL
        ON UPDATB CASCADB,

CONSTRAINT   FK_product_category
        FORBIGN KBY (category_id)
        RBFBRBNCBS category (category_id)
        ON DBLBTB SBT NULL
        ON UPDATB CASCADB
```

6) ORDBR :-

```
CRBATB TABLB ORDBRS (
        order_id PRIMARY KBY,
        customer_id INT NOT NULL,
        order_date DATBTIMB NOT NULL,
        status VARCHAR(30) NOT NULL,
        total_amount DBCIMAL(10,2) NOT NULL,

CONSTRAINT   chk_order_amount
        CHBCK (total amount >= 0),

CONSTRAINT   fk_order_customer
        FORBIGN KBY (customer_id)
        RBFBRBNCBS Customer (customer_id)
        ON DBLBTB CASCADB
        ON UPDATB CASCADB
```

7) ORDER ITEM :-

```
CREATE TABLE OrderItem (
    order_id INT NOT NULL,
    product_id INT NOT NULL,
    price DECIMAL (10, 2) NOT NULL,
    Quantity INT NOT NULL,
    PRIMARY KEY (order_id, product_id)
    CONSTRAINT chk_order_item_quantity
        CHECK (quantity > 0),
    CONSTRAINT fk_odered_item_order
        FORBIGN KEY (order_id)
        REFERENCES Orders (order_id)
        ON DELETE CASCADE,
        ON UPDATE CASCADE,
    CONSTRAINT fk_oder_item_product
        FORBIGN KEY (product_id)
        REFERENCES Product (product_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);
```

8) PAYMENT :-

```
CREATE TABLE Payment (
    payment_id PRIMARY KEY,
    order_id INT NOT NULL UNIQUE,
    transaction_id INT NOT NULL,
    paymentdate DATETIME NOT NULL,
    payment_amount DECIMAL (10, 2) NOT NULL,
    payment_method VARCHAR (30) NOT NULL,
    payment_status VARCHAR (30), NOT NULL,
```

```
CONSTRAINT    chk_payment_amount
              CHECK (amount >= 0),

CONSTRAINT    fk_payment_order
              FOREIGN KEY (order_id)
              REFERENCES orders (order_id)
               ON DELETE CASCADE
               ON UPDATE CASCADE
```

9) DELIVERY :-

```
CREATE TABLE Delivery (
   delivery_id INT PRIMARY KEY
   order_id INT NOT NULL UNIQUE
   address_id INT,
   delivery_date DATETIME,
   delivery_status VARCHAR(30) NOT NULL,
   tracking_no VARCHAR UNIQUE
   delivery_partner VARCHAR(100),

  CONSTRAINT   fk_delivery_order
               FOREIGN KEY (order_id),
               REFERENCES orders (order_id)
                ON DELETE CASCADE
                ON UPDATE CASCADE,

  CONSTRAINT   fk_delivery_address
               FOREIGN KEY (address_id)
               REFERENCES Address (address_id)
               ON DELETE SET NULL
               ON UPDATE CASCADE

);
```

Inserting the values :-

```
INSERT INTO customer ( customer_id , name , email ,
    phone , registered_on) VALUES
    (1, 'Alex', 'alex11@gmail.com', '9871670281',
    '2026-01-10  10:30:00'),
    (2, 'Aman verma', 'amanverma12@gmail.com', '98167-
    54321', '2026-02-10  11:00:00');

INSERT INTO Address ( address_id , customer_id ,
    address_type , street , city , state , pincode , country)
    VALUES
    (101 , 'Home', 'Sector 15', 'Chandigarh', 'Punjab',
    '160015', 'India'),
    (102 , 2, 'Office', 'IT Park Road', 'Chandigarh',
    'Punjab', '160017', 'India');

INSERT INTO Seller ( seller_id , seller_name, email)
VALUES (201, 'Techworld', 'techw@gmail.com'),
(202 , 'FashionHub', 'fashionhub@gmail.com')

INSERT INTO category ( category_id , category_name,
    description ) VALUES
    (301 , 'Electronics', 'Electronic devices and accessories'),
    (302, 'Fashion', 'Clothing and fashion products');

INSERT INTO Product
    ( product_id, product_name, price, seller_id,
    category_id ) VALUES
```

(401, 'Laptop', 65000.00, 201, 301)
(402, 'Wireless Mouse', 1200.00, 201, 301);

INSERT INTO Orders
(order_id, customer_id, order_date, status, total_amount) VALUES
(501, '2026-08-20 10:00:00', 'Delivered', 66,200.00),
(502, '2026-08-21 10:30:00', 'Processing', 2499.00);

INSERT INTO Order Item
(order_id, product_id, price, quantity) VALUES
(501, 401, 65000.00, 1),
(501, 402, 1200.00, 1);

INSERT INTO Payment
(payment_id, order_id, transaction_id, payment_date, amount, payment_method, payment_status) VALUES
(601, 501, 'TXN1001', '2026-08-21 10:05:00', 66200.00) 'UPI', 'Success');

INSERT INTO DELIVERY
(delivery_id, order_id, address_id, delivery_date, delivery_status, tracking_no, delivery_partner) VALUES
(701, 501, 101, '2026-08-22 15:00:00', 'Delivered', 'TKN1001', 'Bluedart');

Demonstrating Relational Integrity Violation

INSERT VALUES INTO Orders
(order_id, customer_id, order_date, status, totalamount)
VALUES

Expected Output :-

ERROR 1452 (2300) : Cannot add or update a child row : a foreign key constraint fails.

Expected Output :-

ERROR 1062 (23000) : Duplicate entry 'alen11@gmail.com' for key 'customer email'.

Expected Output :-

ERROR 1048 (23000) : Coloumn 'email' cannot be null.

(504, 999, NOW(), 'processing', 5000.00);

```
INSERT INTO Customer
(customer_id, name, email, phone, registered_on)
VALUES
(4, 'Rahul Kumar', 'alen11@gmail.com', '9876173217',
NOW());
```

```
INSERT INTO Customer
(customer_id, name, email, phone, registered_on)
VALUES
(4, 'Rahul Kumar', NULL, '9876145732', NOW());
```

OBSERVATION :- The relational schema was sucessfully implemented in MYSQL using constraints. Sample records were inserted sucessfully while invalid and duplicate values were rejected by MYSQL. Referential integrity violations demonstrated that foreign keys present invalid references, UNIQUE prevents duplicate values and NOT NULL prevents missing mandatory values. The behaviour of CASCADE and SET NULL was also verified.

RESULT :- The ecommerce diagram was sucessfully converted into relational table with appropiate keys and integrity constraints. Sample data and constraints violations were sucessfully demonstrated, verifying essential and domain integrity.